Project: *Deep Learning Based Arabic Audio Understanding and Retrieval System*

Intelligent Audio Analysis System (Speech → Text)

1. Project Overview

The goal of this project is to build a **deep learning system that converts speech to text**.

This type of system is used in:

- Meeting assistants
- Lecture transcription
- Podcast search engines
- Call center analytics
- Voice assistants

Students will design a pipeline using neural networks to process audio and extract useful information.

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2. System Pipeline

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Audio Input
   |
   ▼
Speech Recognition (ASR)
   |
   ▼
Text Transcript
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3. Main Tasks

Task 1 — Speech-to-Text

-Convert audio to text use CNN, and LSTM networks combined

- Convert audio to text using popular neural models is suitable for Arabic? find state of the art or any suitable models

Models such as : OpenAI Whisper, Wav2Vec 2.0, and DeepSpeech

Dataset examples: Arabic datasets

1. Mozilla Common Voice (Arabic)
   https://datacollective.mozillafoundation.org/datasets/cmj8u3os6000tnxxb169x1zdc
   https://commonvoice.mozilla.org/en
2. Arabic Speech Corpus
   https://en.arabicspeechcorpus.com/

3. MASC Arabic Speech Dataset
   https://huggingface.co/datasets/hirundo-io/MASC

4. Arabic Broadcast News Dataset
   https://catalog.ldc.upenn.edu/LDC2006S46

5. EJUST dataset (don't share online)

https://drive.google.com/drive/folders/0B28Rhpdi0XLMfldKUUN6ZlhCTmRwLUhaUTZBN3ZBMUFDcjRVWVV3TTFtVTZobUVhMFBGSzg?resourcekey=0-x-kF6vPagWEyH9kjdVSl_g

.Recommended Dataset Combination for the Project

Speech recognition
→ Mozilla Common Voice Arabic

6.Optional Advanced Tasks

- **Intelligent Audio Analysis System (Speech → Text → Summary → Search)**
- **Speaker Identification -** Detect different speakers.

- **Emotion Detection :** Detect emotions in voice. Classes: happy, angry, neutral, sad
- **Keyword Spotting** Detect specific keywords such as: "emergency", "deadline", "exam"

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7. Evaluation Metrics

For Speech Recognition find suitable metrics such as Word Error Rate (WER)

8. Expected Deliverables

Students should submit:

1. Source code
2. Dataset description
3. System architecture diagram
4. Experiments
5. Evaluation results
6. Demo interface

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9. Example Applications

- Lecture search engine
- Podcast summarizer
- Meeting assistant
- Call center analysis